



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDERGRADUATE STUDIES DIVISION

First Semester 2025-26

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : PHA G537

Course Title : PARENTERAL PRODUCT DEVELOPMENT

Instructor-in-charge : ANIL JINDAL

1. **Course description:** B Introduction to parenteral product & their types, products characteristics; Vehicles requirement for Parenteral Products manufacturing; Formulation & process consideration for development of parenteral products; Introduction & understanding of various aspects of unit operation in parenteral product manufacturing; Critical Quality attributes of Parenteral Product; Various Sterilization techniques for parenteral products; Understanding of key requirements for aseptic processing; Understanding of Lyophilization techniques of stable product development; Regulatory consideration for safety assessment of Parenteral Products; Packaging requirement for Parenteral Drug Products; Regulatory perspective for Complex Injectable products development and filing.
2. **Scope and Objective:** This course will provide overview on parenteral product development & designed for students to understand critical aspect of parenteral product development and manufacturing. This course covers basic information, understanding and application of key operations followed during parenteral product manufacturing like lyophilization, aseptic processing, & filtration along with Quality Assurance, control & packaging requirement. This course also covers regulatory consideration for parenteral product assessment with respect to safety for intended application.
3. **Course learning outcomes:**
 - a. Complete understanding of formulation, process, manufacturing and packaging aspect of parenteral drug product
 - b. Complete understanding, importance & application of lyophilization process.
 - c. Knowledge of critical quality attributes in parenteral product development.
 - d. Knowledge of regulatory expectation for safety assessment of parenteral products

4. Textbooks and references:

TB1. Niazi SK, "Handbook of Pharmaceutical Manufacturing Formulations – Sterile Products" Informa healthcare, New York, 2nd Edn, Vol. 6, 2009.

TB2. Akers M.J., Sterile Drug Products: Formulation, Packaging, Manufacturing and Quality, CRC Press, 1st Edition, 2010.

RB1. Sandeep Nema, John D. Ludwig, "Pharmaceutical Dosage Forms: Parenteral Medications", Vol. 1-3, CRC Press, 3rd Edition, 2010.

RB2. Adeboye Adejare "Remington: The Science and Practice of Pharmacy" Pharmaceutical Press, 23rd Edition, 2020.

RB3. Parenteral Drug Association (PDA) Technical Reports.

RB2. U.S. Pharmacopeia, U.S. Pharmacopeial Convention Inc., Rockville, MD, 43rd Edn., 2020.

5. Course Plan: weekly schedule for 14 weeks with [3] lectures per week

Lecture No.	Learning Objectives	Topic to be covered	Ref
1-2	Parenteral Drug product	Introduction, Historical Perspective, Characteristics & Types	T2 Ch 1-3 R1(V1) Ch1
3	Vehicles for Parenteral Products	• Aqueous - Key requirement for High Quality water system (WFI) for parenteral product • Non-aqueous solvents	T2 Ch 6 R1(V1) Ch5
4-6	Formulation & process development of parenteral products	• Formulation development - Selection & optimization of various excipients like Buffers, antioxidants, surfactants, pH adjusting agents, tonicity modifiers, etc. to obtain desired quality attributes • Process development – Key consideration for process development and control of parameters like temperature, light, oxygen, hydrolysis, etc.	T2 Ch 5, 6, 8 R1(V1) Ch 4, 5



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7-9	Unit Operations in Parenteral Product Manufacturing	□ Introduction & understanding of various aspects of parenteral product manufacturing unit operations like dispensing, compounding, filling, stoppering, sealing, inspection, labelling & packaging	T2 Ch 12 R1(V2) Ch1
10-15	Critical Quality attributes of Parenteral Product	• Introduction, understanding & compendial requirement for various quality attributes like • Physical – Surface Tension, Viscosity, Particulate Matter, Color, Clarity, etc. • Chemical – Assay, Impurities, content of functional excipients, etc. • Microbiological – Bioburden, BET, Sterility, CCIT, etc.	T2 Ch 27, 28, 29, 30 R1(V2) Ch5, 6, 7 R1(V3) Ch9
16-24	Sterilization techniques for parenteral products	• Selection of sterilization method – Decision Tree • Terminal Sterilization – Dry Heat and moist heat Sterilization, Cycle development, D-value determination, validation, Indicator, Minimum Maximum load - Filtration – Types of filters like Cartridge, Capsule, Depth, Membrane filters, etc.; Filter Material of Construction like PES, PVDF, Nylon, PTFE, etc.; Modes of filtration – Series filtration, Parallel filtration, etc.; Filter validation studies. - Basic understanding and application of other techniques like Ethylene oxid sterilization, Gamma Sterilization, E-beam sterilization, X-ray sterilization, etc.	T2 Ch 17, 18 R1(V2) Ch8, 9, 10, 11, 12, 13 T1 Ch 4

25-29	Aseptic processing	☐ Understanding of key requirements for aseptic processing like ☐ Personnel, Equipment and building requirement ☐ Cleaning, disinfection & sterilization requirement – CIP, SIP, VHP, etc. ☐ Media fill trials – overview and Approaches	T2 Ch 21 T1 Ch 4
30-33	Lyophilization of Parenterals	☐ Introduction – Principle ☐ Application of Freeze Drying ☐ Formulation aspects of freeze dried products ☐ Development of Lyophilization Cycle and Controls ☐ Key challenges and defects in lyophilized product ☐ Design and Key components lyophilizer	T2 Ch 20 R1(V2) Ch15 USFDA Guidelines Class notes
34-36	Regulatory consideration for safety assessment of Parenteral Products	☐ Understanding of various tests needs to be done to assess safety of parenteral products like ☐ Extractable and Leachable Study ☐ Elemental impurity assessment ☐ Glass Delamination Study ☐ Shelf life assessment using stability Studies	R1(V3) Ch10 T2 Ch 6 USP Chapters
37-39	Packaging requirement for Parenteral Drug Products	Introduction to various type of packaging system for parenteral drug product Innovation in Drug Device Combinations - Prefilled syringes, Pen devices, cartridges system, etc.	T2 Ch 6 R1(V1) Ch11, 12, 13 R1(V3) Ch5 Class notes
40-42	Regulatory perspective for Complex Injectable products	Regulatory consideration with respect to design, development and filing strategy for complex injectable products like Nanoparticles, liposomes, microspheres, nanosuspensions, etc.	Class notes



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6. Evaluation Scheme:

(Given below are suggestions for possible kinds of evaluation components. Indicative only.)

Component	Weightage	Duration	Date and time	Comments
Mid-semester Test	[30%]	[90 min]	As announced in timetable	[Open notes/Closed book]
Quizzes/Seminars/Assignments	[30%]	-	-	-
Comprehensive exam	[40%]	[120 min]	As announced in timetable	[Open notes/Closed book]

Criterion for NC: NC report may be awarded NC when a student does not provide adequate opportunity for evaluation in a course. This could include not submitting required assignments, not attending mandatory sessions, or failing to actively participate in course activities. If the overall performance in assessments is deemed insufficient for evaluation, an NC may be assigned.

7. Chamber consultation hours: As announced in the class

8. Makeup Policy: A student may apply for the makeup if he/she is not able to attend regular midsemester and comprehensive exams due to genuine exigencies. No makeup request will be accepted for quizzes and other continuous assessment component.

[signature]

Instructor-in-charge

[Course Number]